

IoT-Enabled Autonomous Surveillance Robot for Industrial Hazard Detection Using LiDAR-
Based Navigation and Multi-Sensor Fusion

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Abstract

Industrial working environments rely mostly on risky setting that may comprise gas emission, fire breakouts and overheating machines. The traditional systems of monitoring are based on fixed sensors and human observation and inspection that, very often, prevents timely detection and covering of industrial sites.

To overcome those drawbacks, in this study, an IoT-based Autonomous Surveillance Robot with the capability to scout industrial areas and identify the presence of hazards in the environment in real-time became the proposed solution. The system combines the Raspberry Pi 5 as the primary processing unit and several environmental sensors such as MQ2 gas sensors, DHT11 temperature sensors, and humidity sensors, and flame detection modules. To navigate, the robot uses the Simultaneous Localization and Mapping (SLAM) which is LiDAR-based to cover industrial areas without hitting any objects.

The sensor data and the live video feed are sent to the monitoring dashboard in the cloud via an IoT communication (MQTT protocol) framework and data collection. In case dangerous circumstances are realized, SMS and email notifications to operators are automatic.

As a part of experimental evaluation, its evaluation of 150 trials proves that the proposed is correct.

System has a hazard detection accuracy of about 96 percent, but at the same time, the system retains a dependable autonomous navigation and coverage of the area that is being monitored. The system drastically eliminates the necessity of human operators in hazardous industrial areas and provides a cost-effective method of enhancing the level of industrial security inspection.

Keywords: IoT surveillance robot, industrial surveillance, autonomous robots, LiDAR SLAM, hazard detection, smart factor.

1. Introduction

The fast development of Industry 4.0 technologies has greatly altered the current way of manufacturing process because it introduced automation, real-time information exchange, and smart monitoring tools to industrial processes. The advancements have enhanced productivity, efficiency of operations and decision making in different industries. Repin, however, with these technological advances, workplace safety has still remained a big challenge in most industrial

settings. On manufacturing plants, chemical processing, and warehouses often reside dangerous materials, burning gases, high temperatures machinery, and complicated working systems which can be a very dangerous hazard unless observed at all times. Inability to address such risks at the initial stage may result into industrial accidents, equipment loss, environmental damages, and even threats to the human life.

The classical industrial control systems usually depend on the hard-wired sensor constructions and the manual control of safety personnel. Although these methods are considered to offer a simple form of monitoring, they are also associated with a number of critical challenges. The networks of the static sensors are typically mounted in fixed locations in a facility, and this may cause blind zones or coverage gaps in which dangerous conditions can arise without being noticed. Moreover, human lives can also put workers to hazardous conditions as the workers having human personnel assist in regular inspections or safety checks or where the temperature is destabilizing. The human monitoring systems also experience fatigue, slowness in response and errors in working and this may decrease their efficiency in the critical scenarios.

Recent years saw the beginning of new possibilities in the domain of robotics and Internet of Things (IoT) technologies that allow one to create intelligent surveillance systems that are able to monitor autonomously. Incorporating mobile robotic platforms with environmental sensors and wireless communication technologies, combined with processing of acquired data into the cloud, one can design systems to continually survey the industrial facilities and identify real-time dangerous situations. The autonomous robotic system has the capability to travel in various regions of a plant, gather environmental information, and send it to the far monitoring systems, hence decreasing the use of manual inspection and enhancing the safety system structure in general when dealing with industrial workplaces.

This study aims at designing an IoT-based mobile surveillance robot that is able to identify industrial hazards on time. The suggested system will be a unification of autonomous navigation, multi-sensor environmental inspection, and cloud-based communication technologies and implement an all-encompassing industrial safety surveillance. It is proposed that a combination of LiDAR-based navigation system with environmental sensing and IoT communication will help ensure the continuous monitoring of the industrial environment, whilst reducing the level of human exposure to potentially dangerous areas. The suggested solution helps to develop smarter, safer, and more efficient industrial monitoring systems in contemporary Industry 4.0 setting.

2. Related Work

A number of authors have already delved into applications of Internet of Things (IoT) technologies in conjunction with robotic surveillance systems to help monitor and manage safety in industries. The IoT-powered monitoring solutions help collect data of different sensors in real time and access environmental data remotely and, thus, help industries to identify unsafe conditions more effectively. Nevertheless, a significant number of them continue to be based on stationary sensor nodes, which restricts their capabilities when it comes to the monitoring process of large and variably evolving industrial environments.

Vardhan et al. (2023) built an IoT-based monitoring system that can gather information about the environment by means of wireless sensor networks. Their strategy was to keep track of the environmental conditions like temperature and gas concentration in the environment, using sensor nodes distributed. Although the system showed good data acquisition and remote monitoring, it was completely based on the deployment of stationary sensors. This made the system have a limitation on its spatial coverage and the inability to capture hazards beyond the set sensor locations which has the potential of creating blind spots in giant industrial centres.

Xu et al. (2024) suggested a surveillance robot as an autonomous vehicle that will work at construction plants and dynamic outdoor locations. Their system made use of robotic movement and wireless communication to acquire visual data as well as being able to watch what was transpiring in the activities. The robot was effective in terms of movement and monitoring, but the system was largely concentrated on visual surveillance, and it did not have sensors that would detect environmental hazards e.g. gas sensors, flame sensors or temperature sensors. This means that a number of important industrial hazards, which necessitate technologically advanced environmental sensing, were not recognizable by the robot.

Likewise, Bhuiyan et al. (2023) developed a multipurpose surveillance robot with cameras and obstacle sensors that were used to keep track of the environments around. The robot also showed enhanced mobility and a capability of carrying out a few surveillance functions. The system was however primarily founded on visual surveillance and basic obstacle evading mechanisms. It lacked sophisticated sensor fusion capabilities and environmental hazard detection features that could be used to provide an all-inclusive safety monitoring of factories.

Xu et al. (2024) tried to suggest an autonomous surveillance robot that was to work at construction sites and at dynamic outdoor locations. To gather visual data and track the activities being done, their system was supported by the usage of robots with mobility and wireless communication. The robot was shown to have good movement and monitoring behavior, but the system was mainly concerned with visual surveillance, and no gaseous, flame, or temperature sensors were integrated environmental hazard sensors. As a result, the robot could not detect

some of the most important industrial risks that this necessitates particular environmental sensing.

In the same regard, Bhuiyan et al. (2023) came up with a multifunctional surveillance robot that has cameras and obstacle detection sensors to survey the environment around it. The robot showed enhanced movements and capability to carry out simple surveillance activities. Nevertheless, the system depended predominantly on the visual control and basic obstacles avoidance means. It failed to implement advanced sensor fusion methods and environmental hazard identification skills to provide full industrial safety monitoring facilities.

The recent developments in edge computing systems like Raspberry Pi have helped enhance the realization of the idea of smart robotic monitoring systems. Edge computing enables sensor data to be computed at the embedded level and therefore provides online results and less dependence on cloud-based computing. The method enhances responsiveness of the system and lowers network latency when it comes to real-time monitoring applications. However, following these technological transformations, there are no unified solutions, which integrate autonomous robots, multi- sensors ways of environmental hazards, and an IoT-driven alerting into one robotic platform.

The gap presented in the proposed research is justified by the fact that it proposes a particular implementation of the IoT-enabled mobile surveillance robot that will be capable of autonomous navigation and real-time surveillance of hazards occurring in an industrial setting. The proposed system will incorporate the LiDAR-powered navigation, the multi-sensed environmental detection, and the cloud-based warning mechanisms that will enable the suggested solution to be a comprehensive and scalable solution to the problem of industrial safety monitoring.

Table 2.1 Comparison with existing systems

Feature	Vardhan et al. (2023)	Bhuiyan et al. (2023)	Xu et al. (2024)	Proposed System
Autonomous navigation	No	Limited	Yes	Yes
Environmental sensors	Yes	No	No	Yes
Multi-sensor fusion	No	No	No	Yes
Real-time video monitoring	No	Yes	Yes	Yes
Hazard detection capability	Limited	No	No	Yes

Feature	Vardhan et al. (2023)	Bhuiyan et al. (2023)	Xu et al. (2024)	Proposed System
IoT cloud alerts	Yes	No	Limited	Yes

3. System Architecture

The suggested system architecture comprises of four significant modules, which are sensing, processing, navigation, and communication.

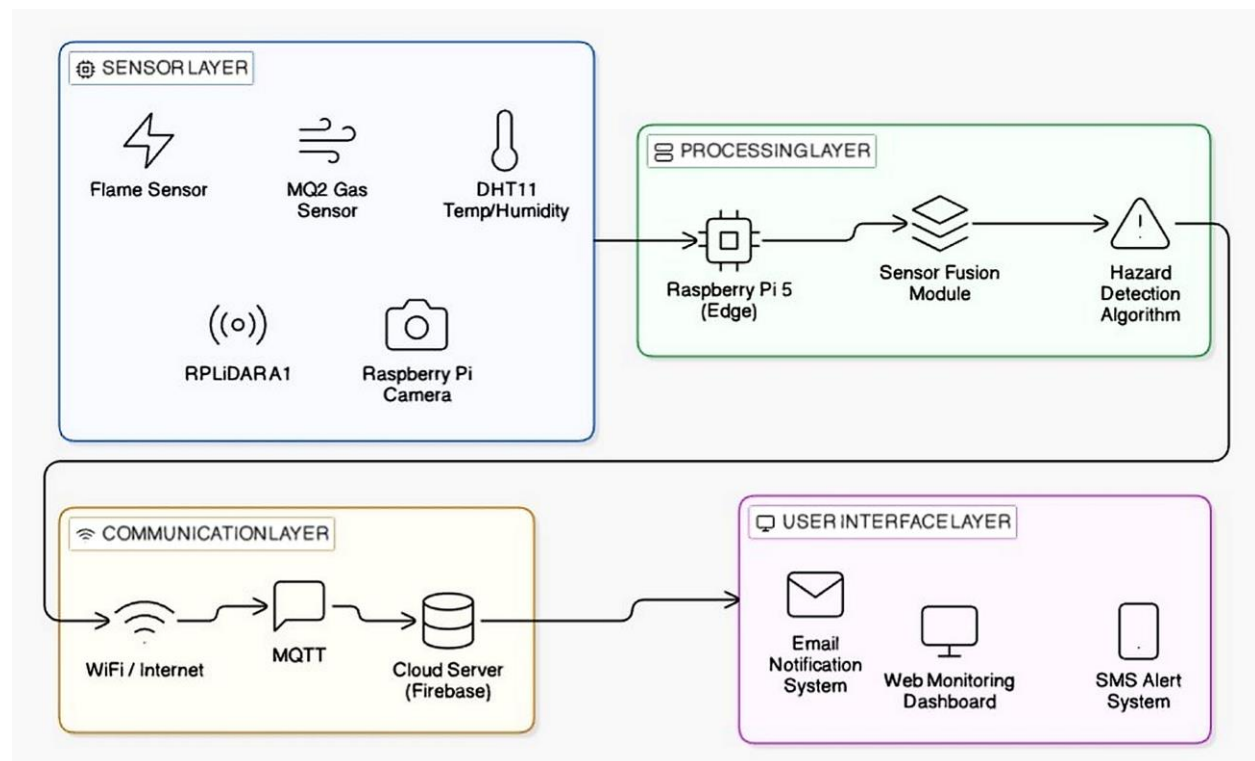


Fig 3.1 System architecture of the IoT-enabled surveillance robot

Hardware Architecture

The main computing power of the robot is the Raspberry Pi 5 one-board computer. It manages the process of sensor data collection, navigation algorithms and connections with cloud servers. The robot will incorporate several sensors that will be able to detect various instances of gas

combustion, temperature and humidity, and fire hazards, which are some of the sensors incorporated by the robot, such as MQ2 gas sensors, DHT11 sensors, and flame sensors respectively. A Raspberry Pi camera module is used to give real-time visual monitoring of the area under monitoring.

Table 3.1 Hardware & sensor specifications

Component	Function	Operating Specification	Range
Raspberry Pi 5	Main processing unit and system controller	2.4 GHz processor	quad-core
MQ2 Gas Sensor	Detect combustible gases such as LPG, methane, and smoke	300–10000 ppm	
DHT11 Sensor	Measures temperature and humidity	Temperature accuracy $\pm 1-2^{\circ}\text{C}$	
Flame Sensor	Detects infrared radiation from fire sources	Detection range up to 3 m	
RPLiDAR A1	Environmental scanning and navigation mapping	360° scanning, up to 12 m range	
Raspberry Pi Camera	Real-time visual monitoring	1080p video resolution	
DC Motors	Robot locomotion and movement	12V high torque motors	

Navigation System

They involve Autonomous navigation with the help of an RPLiDAR A1 sensor. The LiDAR sensor conducts an environmental scanning and creates a 2D map with the Simultaneous Localization and Mapping (SLAM) algorithms. This will allow the robot to manoeuvre within industrial settings without colliding with the obstacles.

Sensor Fusion

The system is based on sensor fusion to enhance reliability and minimize the need to trigger false alarms. They do not rely on a single sensor reading, but rather the system is used to check the

hazardous conditions by integrating multiple sensor readings. As an example, a rise in the level of gases and increase in temperature is a sign of possible combustion.

IoT Communication

The robot also interacts with a cloud-based monitoring system via the MQTT protocols. Sensor data and video stream are sent to a web-based dashboard where operators are able to monitor the environment remotely. In the case of detection of hazardous conditions, automated alerts are transferred through SMS and emails.

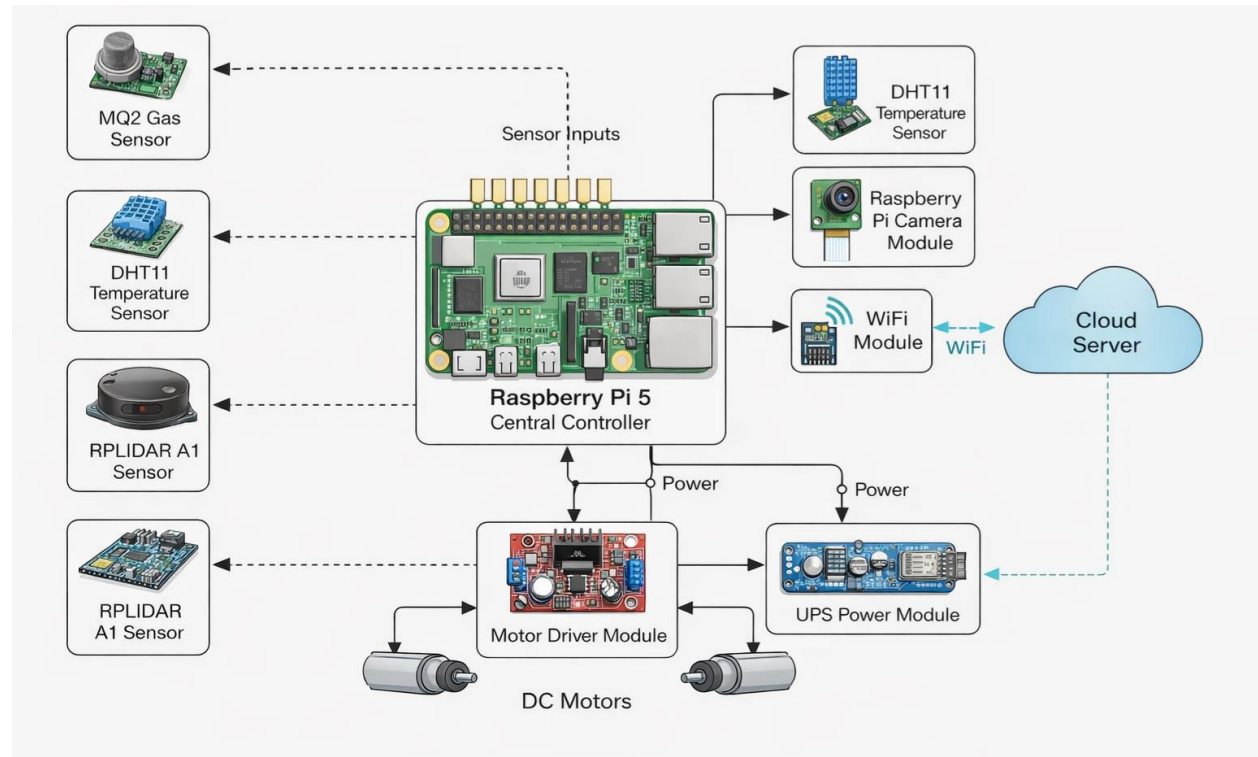


Fig 3 2. Hardware block diagram of the surveillance robot

4. Methodology

The proposed system of managing surveillance robots was developed based on Python programming language and embedded in the Robot Operating System (ROS) platform. ROS is based on a modular and flexible structure that can be easily tailored to fit a variety of subsystems such as navigation algorithms, sensor drivers and communication protocols. The system can handle the data exchange between hardware components including environmental sensors,

LiDAR modules and motor control units using ROS nodes and topics. The Python language was chosen because it has a wide range of support of robotics libraries and Ros packages which make it easier to develop sensor processing, hazard detection algorithms and also the communication interfaces.

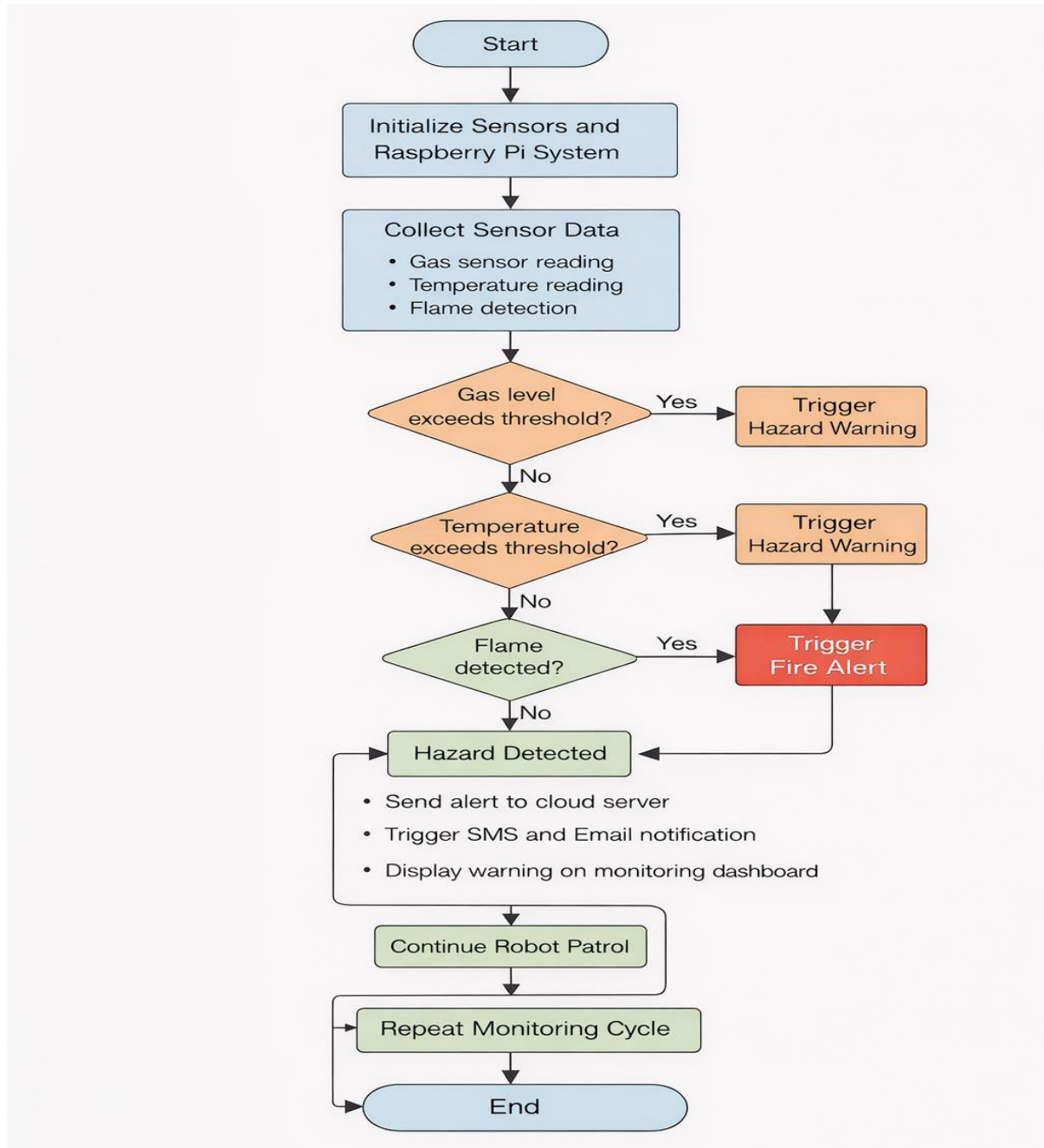


Fig 4.1 Hazard detection algorithm flowchart

The mechanism of the robot is that predefined patrol patterns are adhered to in the industrials to allow systematic inspections of dangerous areas. The data of the environment, collected by the integrated sensors on the robot (MQ2 gas sensor, DHT11 temperature and humidity sensor, and flame detection module), is activated at each patrol cycle. The sensor values gathered are analyzed locally on the Raspberry PI controller based on predefined threshold values in order to detect abnormal conditions in the surrounding. To enhance the reliability of the detection, sensor fusion method is used to verify the existence of hazardous conditions by the measurements of multiple sensors in lieu of a single sensor. When sensor readings have surpassed the stipulated safety levels, the system automatically activates an alarmed system that makes notifications to both the remote monitoring dashboard and SMS and email alerts to the authorized individuals.

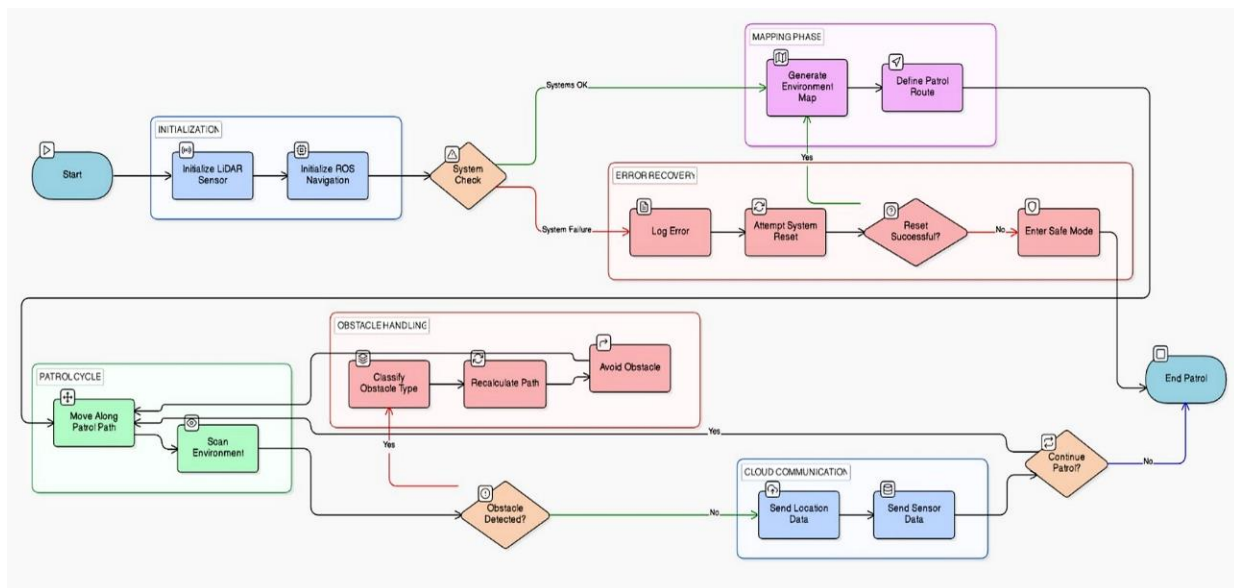


Fig 4.2 Autonomous navigation and patrol workflow

In order to test the efficacy of the proposed system, the experiments were done in two environments. The initial testing condition was a controlled laboratory and sensor calibration and the functionality of the system was checked under the conditions of stable operation. The second setting was a pseudo-industrial working environment which included equipment, physical barriers, and the different environmental factors. This configuration enabled the system to be put to test on dynamic as well as realistic operating conditions. The number of experimental trials was 150, which allowed studying the performance of a system in relation to three criteria: accuracy of detecting hazards, reliability in navigation, and the time of reaction to an alert. These

experiments allowed to prove that the robot can identify dangerous situations and that it is able to stay stable on autonomous movement in complicated conditions.

5. Experimental Results

Experimental testing was aimed at the accuracy of hazard detection, reliability of the navigation, and the time required by the system.

Hazard Detection Accuracy

The level of the MQ2 gas sensor was set at 94.5 percent of the average detection accuracy at a radius of 2.5 meters. The flame detection module had an accuracy of 97.1 and the DHT11 temperature sensor had a variation of ± 1.2 O C compared to industrial thermometers. The total accuracy of the hazard detection was 96%.

Autonomous Navigation

The robot was capable of planning on industrial environments by using LiDAR-based algorithms of SLAM to create correct maps of the environment. The robot covered around 98 percent of the mapping accuracy and avoided moving barriers during the majority of the trials of the experiment.

System Latency

The average duration that was taken to detect the hazard and transmit the alert was 1.7 seconds. This reaction time is much quicker as compared to the conventional human-permitted methods of checking.

Table 2.1 Experimental performance results

Performance Parameter	Result
Gas detection accuracy	94.5%
Flame detection accuracy	97.1%
Overall hazard detection accuracy	96%
SLAM mapping accuracy	98%

Performance Parameter	Result
Obstacle avoidance success rate	95%
Average alert response time	1.7 seconds
Total experimental trials conducted	150

6. Discussion

The experimental findings prove that the combination of Internet of Things (IoT)-technologies and autonomous robotic platforms can contribute greatly to the industrial safety surveillance system. The surveillance robot proposed has been able to integrate environmental sensing, autonomous-navigating, and cloud-based communications, to build an all-comprising monitoring system in industrial environments. The system can detect different dangerous conditions and in real time by incorporating several sensors, e.g., gas detectors, temperature sensors, and flame detectors. The navigation based on LiDAR can make the robot move independently in the complicated industrial plans to prevent collisions, which makes the robot constantly monitor the area without necessarily involving people.

The proposed mobile robotic platform is more flexible and extensive in comparison with the traditional monitoring systems that require the use of non-portable sensors. The use of the fixed surveillance systems is prone to have fewer fields of coverage and not sense any dangers that fall beyond the range of the sensors. Contrary to this, the portability of the robot can enable the robot to cover the various areas of a facility and dynamically give attention to various areas on every patrol run. This mobility will greatly decrease chances of blind spots and will be able to detect dangerous events at a more advanced point thus enhancing overall safety and efficiency in the operations of industrial institutions.

The other major benefit of the suggested system is the utilization of edge computing functionality offered by the Raspberry Pi processing unit. The robot does not send raw sensor data go to cloud servers to be analyzed, rather analyze the data locally to decide on the presence of abnormal conditions. This has been implemented in this local processing and made sure that the system latency has been minimized and that the response is faster at critical times. The system also conserves bandwidth on networks and increases reliability in conditions where accessibility is problematic, the system can also reduce constant contact between clouds. These features show the usefulness of integrating edge computing with the IoT-based robotic systems in the industrial context of real-time monitoring.

7. Conclusion

This report proposed the architecture and a deployment of an autonomous surveillance robot based on IoT to be used towards the hazards detection both in the industry and environmental surveillance. The suggested system is a mixture of some technologies, including environmental sensors, LiDAR-based navigation, edge computing, and cloud-based communication and creates a system of intelligent monitoring that can track dangerous conditions in real-time. The system has capability of taking sensor data, navigation, and communicating with other surveillance through Raspberry Pi which is the central processor.

This solution is proved to be effective due to the experimental evaluation that was performed in simulated and controlled industrial environments. The robot was very accurate in its Hazard detections, and simultaneously, has ensured autonomous navigation obtained in a fluctuating working environment, which is hazards. Integrating several sensors will improve the improvement in the detecting process and allow the system to identify different types of industrial hazards such as gas leaks, fire outbreak, and anomalous changes in temperatures.

The system that turns out will make a significant contribution to the workplace safety as it reduces the human exposure in the dangerous areas. The real-time alert systems and constant monitoring will ensure that the operators are informed about the dangerous conditions in time and the solution will be available to take a preventive measure and respond in time. In addition, the proposed system is economical and can be scaled and can be applicable in the requirements of the various industry settings which include manufacturing factories, warehouses, chemical plant, and power plants. Overall, the integration of the IoT technologies and autonomous robotic platforms can be considered as one of the potential solutions to the development of smarter and safer industrial surveillance systems.

8. Future Scope

Although the proposed surveillance robot demonstrates possible effectiveness in the industrial hazard monitoring, it is possible to consider a range of modifications that can be made to improve the functionality of the system. One of such improvements can be regarded as the addition of advanced computer vision techniques based on the deep learning algorithms. Light neural networks, e.g., YOLO or MobileNet, helped the robot to recognize smoke, draw the compliance with safety norms, or unwarranted human activity in the factories. Such functions would not only increase the monitoring system to functions of environmental sensor but also be able to perform a visual safety analysis.

The second possible solution is the emergence of the swarm robotics technologies. Instead of a single surveillance robot, a number of robots can be used to collaborate in a large industrial facility. Such robots were able to communicate with each other as well as align the routes which they were covering in a way that they would cover the entire area that they were patrolling. The swarm-based monitoring would substantially improve efficiency and reduction of the time of monitoring as well as tolerance to faults on a large scale in the industrial environment.

In the future, the predictive maintenance system which is implemented using machine learning algorithms can also be introduced. Such analysis of trends and the ability to extract indicators of equipment malfunction or an unusual machine performance would be possible through the constant collection of environmental and operation data. It is so because predictive maintenance models will help industries to prevent accidents even before they occur and minimize the operation downtimes. Furthermore, the additional evolution of the communication infrastructures, such as 5G network and special purpose AI accelerators, could enable the growth of the power of the processing of the system, enabling to conduct the analysis of the data at a faster pace, and making the system more efficient when responding to the sudden industrial operational changes.

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References

Al-Fuqaha, A., Guizani, M., Mohammadi, M., Aledhari, M., & Ayyash, M. (2015).

Internet of Things: A survey on enabling technologies, protocols, and applications. *IEEE Communications Surveys & Tutorials*, 17(4), 2347–2376.

<https://doi.org/10.1109/COMST.2015.2444095>

- Bhuiyan, M. Z., Islam, S., & Rahman, M. (2023). Designing a multifunctional surveillance robot for environmental monitoring. *Global Journal of Engineering and Technology Advances*, 15(1), 20–28.
- Chitale, Y. P., Patil, A., & Deshmukh, S. (2024). IoT-based industrial surveillance and hazard detection robot. *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering*, 13(2), 112–118.
- Durrant-Whyte, H., & Bailey, T. (2006). Simultaneous localization and mapping (SLAM): Part I. *IEEE Robotics & Automation Magazine*, 13(2), 99–110. <https://doi.org/10.1109/MRA.2006.1638022>
- Gupta, R., Sharma, P., & Singh, A. (2022). Smart industrial monitoring system using IoT and wireless sensor networks. *International Journal of Distributed Sensor Networks*, 18(6), 1–10. <https://doi.org/10.1177/15501477221104589>
- Kim, J., Park, S., & Lee, K. (2021). Autonomous mobile robot for real-time industrial surveillance using LiDAR and IoT technologies. *Sensors*, 21(18), 6123. <https://doi.org/10.3390/s21186123>
- Patel, K., Patel, S., & Scholar, R. (2016). Internet of Things (IoT): Definition, characteristics, architecture, enabling technologies, application and future challenges. *International Journal of Engineering Science and Computing*, 6(5), 6122–6131.
- Siciliano, B., & Khatib, O. (2016). *Springer handbook of robotics* (2nd ed.). Springer.
- Song, D., Wang, J., & Zhang, Y. (2023). Design and implementation of a smart surveillance system for environmental monitoring. *Journal of Electrical Engineering & Technology*, 18, 583–592.
- Thrun, S., Burgard, W., & Fox, D. (2005). *Probabilistic robotics*. MIT Press.
- Vardhan, B. S., Reddy, P. K., & Kumar, M. (2023). Autonomous surveillance and monitoring boat using IoT. *International Research Journal of Engineering and Technology*, 10(10), 299–302.
- Xu, Y., Li, Z., & Chen, H. (2024). IoT-based autonomous surveillance robot for monitoring construction sites. *Journal of Sensor and Actuator Networks*, 13(1), 10. <https://doi.org/10.3390/jsan13010010>
- Zhang, T., Wang, H., & Chen, Y. (2022). An IoT-based intelligent monitoring system for industrial safety management. *IEEE Internet of Things Journal*, 9(14), 11823–11833. <https://doi.org/10.1109/JIOT.2021.3138472>